

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
INSTRUCTION DIVISION
Course Handout (Part II)

17.08.2026

In addition to part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No. : BIO G525
Course Title : ENVIRONMENTAL BIOTECHNOLOGY AND WASTE MANAGEMENT
Instructor-in-charge: B. Vani
Instructor: Prof. S.K. Verma

1. Course Description:

Application of biotechnology to the management of environmental problems. Biotechnology for enhanced plant and animal production through biological insecticides, herbicides resistance, mineral cycling, conservation of genetic resources and biological nitrogen fixation. Use of biotechnological processes for pollution control, bioremediation of toxicants and treatment of domestic and industrial wastes. Ethical issues related with the release of genetically modified organism.

2. Scope and Objectives:

The major objective of this course is to impart knowledge on application of biotechnological processes for betterment of environment. This course is designed to make the student understand the various biological phenomena, which can be exploited to save environment from eventual deterioration.

1. Text Book: Environmental Biotechnology by Alan Scragg, Oxford University Press, UK.2015

2. Reference Book

Environmental Biotechnology: Basic concepts and applications by Indu Shekhar Thakur, Second edition. I.K. International Publishing House Pvt. Ltd. New Delhi 2011.

3. Course Plan

Module No.	Lecture Session	Reference	Learning outcomes
1. Introduction and scope of environmental biotechnology	Basic concept of environment and its components. Biotechnology for environment; definitions and facts	Chap. 1, TB + Research papers	To introduce them to the subject of environmental biotechnology and its scope
2. Environmental pollution	Sources of various pollutions and their environmental impact.	Chap.1, TB	
3- 6 Environmental monitoring	Methods for the measurement of pollution. Air, water and soil sampling; Analyses of samples; physical, chemical and biological methods. Nucleic acid based techniques for analyses of diversity, structure and dynamics of microbial	Chap.3, TB + Research papers	Students will gain knowledge on basic pollution monitoring techniques which can be used to get into the field based Govt. jobs or into research on recent strategies to take care of different types of pollutions.

	community in wastewater treatment, Concept of biomarkers. Metagenomics.		
7-9 Biotechnology of sewage treatment	Basics of sewage treatment processes. Function of various treatment systems. Microbiology of sewage treatment	Chap.4 TB	Student will get complete knowledge on functioning of sewage treatment plant and he can apply his knowledge on biotechnology to improvise the methods currently being used.
10 Development of clean technology (minimization of waste generation)	Fundamentals of clean technology. Integrated pest management and bio-control of plant diseases. Microbial polymer production and bio-plastic technology	Chap. 6 TB	The student will learn about alternatives to polythene menace and other harmful chemicals. He can use this knowledge to enter into the area of applied research, which is in great demand.
11-13 Nano-biotechnology and Environment	Pros and cons of nano-biotechnology	Class notes	This topic will make the student competent in the contemporary research/technologies.
14-17 Bioresource technology development	Biotechnology for energy production; basic concept. Biological energy sources and bio-fuels. Biotechnology for enhanced oil recovery	Chap. 7 TB	The student will gain an insight into how to tap various resources and reduce load on fossil fuels.
18 Ethical issues in environmental biotechnology	Release of genetically modified organisms	Chap. 9 TB	This topic will help the student to become a scientist with good morals.
19- 20 Present state of India's environment	Pressing environmental issues in India, Government's policies and programs	Class Notes	This will make the student aware of the pollution status of his own country and, what the Govt. is doing and hence deduce what can be done to improve the present situation.
21-22 Bioremediation of inorganic pollutants (nitrate and phosphate)	Nitrification and denitrification–microbial fundamentals and application. Biological phosphate removal	Chap. 5 TB RB 1	The student can use the knowledge to propose some small scale projects to Govt./non-Govt. funding agencies to suggest novel pollution control mechanisms
23-26 Bioremediation of inorganic pollutants (heavy metals and radionuclides)	Microbial interaction with metallic elements. Metal Toxicity. Molecular mechanism of metal resistance. Biosorption and biotransformation of metals and radionuclides. Recent developments in metal bioremediation.	Chap.5, TB & RB 1, Class notes	The student can use the knowledge to propose some small scale projects to Govt./non-Govt. funding agencies to suggest novel pollution control mechanisms

27-29 Bioremediation of organic pollutants	Aerobic and anaerobic degradation of organic pollutants. Principles, biochemical pathways and genetic regulation. Degradation of aliphatic, aromatic, polyaromatic and chlorinated compounds. Recent developments in waste treatment.	Chap.5 TB RB 1, Class Notes	The student can use the knowledge to propose some small scale projects to Govt./non-Govt. funding agencies to suggest novel pollution control mechanisms
30-31 Phytoremediation	Use of plants for removal of organic and metallic pollutants	Chap. 5 TB RB 1, Class notes	Student can work on new varieties of plants capable of eliminating pollutants.
32-35 Bio-mining of metals and radionuclides	Concepts of bioleaching, microbial aspects, regulatory factors and process application	Chap 8 TB Class Notes	The student will learn how metals can be extracted profitably from low grade ores.
36-37 Disposal and treatment of medical waste	Sources, health hazards, treatment and disposal technologies	Lecture Notes	Awareness of the student towards medical wastes is improved so that he may follow safe practices in future and also enlighten others on various safety issues.
38-40 Biosensor technology	Principle, types and applications	Chap.3 TB, RB 1, Class Notes	Biosensor fabrication is a huge industry and has wide market potential. So. Good job opportunities in biotech sector.

4. Examination Scheme:

component	Duration	Weightage (%)	Date and time	Nature of Component Closed book/open book
Mid Semester	90 min	25	10/10 AN2	CB
Lab Assignment		15		
Quiz (several)	20 Min	10		CB
Seminar	--	10		
Comprehensive Examination	3.0 hrs	40	16/12 AN	CB + OB

5. Chamber Consultation Hour: To be announced

6. Notices: Notices will be displayed on Biological Science Notice Board

7. Make-up Policy: No make-up except in case of hospitalization.

8. Note (if any): None

**Instructor-in-charge
BIO G525**